

DIFFERENTIAL PRIVACY

A Practical Framework for Secure Data Analysis and Strategic Reporting

What is Differential Privacy?

Differential privacy is a rigorous mathematical definition of privacy. In the simplest setting, consider an algorithm that analyzes a dataset and computes statistics about it (such as the data's mean, variance, median, mode, etc.). Such an algorithm is said to be differentially private if by looking at the output, one cannot tell whether any individual's data was included in the original dataset or not.

In other words, the guarantee of a differentially private algorithm is that its behavior hardly changes when a single individual joins or leaves the dataset – anything the algorithm might output on a database containing some individual's information is almost as likely to have come from a database without that individual's information.

Dwork, McSherry, Nissim, and Smith '06.

Privacy as quantifiable measure

A crucial feature of differential privacy is that it defines privacy not as a binary notion of "was the data of individual exposed or not", but rather a matter of accumulative risk. That means, every time a person's data is processed her risk of being exposed increases.

To this end, the definition of differential privacy is equipped with parameters ("epsilon and delta") that quantify the "privacy loss" that is the additional risk to an individual that results from her data being used. Regardless of any auxiliary knowledge used in a re-identification attack, the risk to one's privacy caused by a differentially private algorithm will forever be bounded by this privacy loss.

When is DP helpful?

Through extensive theoretical research, differential privacy shows promise in enabling research data to be shared in a wide variety of settings. The reach of differential privacy extends far beyond the simple case of statistical queries. For instance, there are differentially private versions of algorithms in machine learning, game theory and economic mechanism design, statistical estimation, and streaming.

It is worth remarking that differential privacy works better on larger databases. This is because as the number of individuals in a database grows, the effect of any single individual on a given aggregate statistic diminishes.

Formal Mathematical Definition

Differential Privacy is formally defined as: A randomized algorithm M with domain $\mathbb{N}^{|\mathcal{X}|}$ is (ε, δ) -differentially private if for all $S \subseteq \text{Range}(M)$ and for all $x, y \in \mathbb{N}^{|\mathcal{X}|}$ such that $\|x - y\|_1 \leq 1$:

$$\Pr[M(x) \in S] \leq \exp(\varepsilon) \Pr[M(y) \in S] + \delta$$

The Epsilon (ε) and Delta (δ) parameters measure the threshold for leakage:

- **Epsilon:** Defines how different the actual data is from the queried data. If $\varepsilon = 0$, $\exp(\varepsilon) = 1$ which means both the data are equal.
- **Delta:** Is the probability that an information will accidentally be leaked as compared to the value of Epsilon. If $\delta = 0$, that means no data is being leaked.

When both Epsilon and Delta are 0, it is called *Perfect-Privacy*. The values are set in such a way so that the privacy is maintained. This set of values is known as **Privacy-Budget**.

Roles in our Scenario

In our specific operational context, the roles are defined as follows:

- **Data Generators:** The individuals whose information populates the dataset.
- **Data Owner:** The agency that commissions us to process and analyze the data.
- **Data Curator & Aggregator:** We act as both. We are responsible for safeguarding the information, performing the analysis, and ensuring privacy at the output stage.

How is DP implemented?

This new area of research addresses the paradox of learning nothing about an individual while learning useful information about the population. This is done by sending queries to the data curator. The goal is to answer with highest possible accuracy without leaking information using two main methods:

a. Local Differential Privacy (Local DP)

In local differential privacy the random noise is applied at the start of the process (local level) i.e. when the data is sent to the data curator/aggregator. If the data is too confidential, generally the data generators do not want to trust the curator and hence add noise to the dataset beforehand. This is adopted when the Data Curator cannot be completely trusted. The main cons is the loss of accuracy of analysis result.

b. Global Differential Privacy (Global DP)

In Global differential privacy the random noise is applied at the global level i.e. when the answer to a query is returned to the User. This type of differential privacy is adopted when the Data generators trusts the data curator completely and leaves it to the curator the amount of noise to be added to the results. This type of privacy results is more accurate as it involves lesser noise. The advantage is that global dp does not influence given data and secure reliable analysis results.

Thoughts on DP for 4M

During our research into privacy-enhancing algorithms, we identified Differential Privacy (DP) as the most reliable and least invasive technique. While DP is an essential tool for safeguarding client data, it introduces statistical noise that can result in slight data variations, potentially affecting **LLM interpretations** and analytical accuracy.

In my view, DP should not be applied universally across all 26 domains, as much of our data does not contain sensitive information. While implementing DP would be a significant asset for **4M**, our primary challenge lies in **automating the classification process** to determine which datasets require this algorithm and which do not.

INDUSTRY STANDARDS AND LIBRARIES

Top tech companies like **Apple, Google, and Uber** use DP daily. Differentially Private Data Analysis is implemented by using various libraries:

- **PyDP** (OpenMined): A Python Wrapper for Differential Privacy.
- **OpenDp** (Harvard and Microsoft).
- **Diffprivlib** (IBM).
- **Google's Differential Privacy Library**.